

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406351
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kitano; Shinya et al.

Video confirmation computer

Abstract

A video confirmation computer for confirming video related to robot operation includes a storage unit and a processor. The storage unit stores information on the position of the electric motor that drives a link body of the robot received from the controller of the robot and information on the video obtained by a camera attached to the robot. The processor makes at least one of the video confirmation computer itself and a computer connected to the video confirmation computer display a model area and a video area side by side. In the model area, a two-dimensional or three-dimensional model reproducing the posture of the robot is displayed by computer graphics. In the video area, the video is displayed.

Inventors: Kitano; Shinya (Kobe, JP), Nakaya; Atsushi (Kobe, JP), Yoshida; Masaya (Kobe, JP), Fujimori; Kazuo (Kobe, JP), Okada; Hiroyuki (Kobe, JP)

Applicant: KAWASAKI JUKOGYO KABUSHIKI KAISHA (Kobe, JP)

Family ID: 76311372

Assignee: KAWASAKI JUKOGYO KABUSHIKI KAISHA (Kobe, JP)

Appl. No.: 18/754454

Filed: June 26, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240346641 A1	Oct. 17, 2024

Foreign Application Priority Data

JP	2019-222900	Dec. 10, 2019
JP	2020-099458	Jun. 08, 2020

Related U.S. Application Data

continuation parent-doc US 17783297 US 12045972 WO PCT/JP2020/045186 20201204 child-doc US 18754454

Publication Classification

Int. Cl.: G06T7/00 (20170101); G06F3/04847 (20220101); G06T11/00 (20060101)

U.S. Cl.:

CPC G06T7/0004 (20130101); G06F3/04847 (20130101); G06T11/00 (20130101); G06T2207/30148 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11110599	12/2020	Goto et al.	N/A	N/A
11203121	12/2020	Goto et al.	N/A	N/A
11335578	12/2021	Yoshida et al.	N/A	N/A
11476138	12/2021	Goto et al.	N/A	N/A
12045972	12/2023	Kitano et al.	N/A	N/A
2007/0144673	12/2006	Yeom	156/345.24	C23C 16/54
2014/0068439	12/2013	Lacaze et al.	N/A	N/A
2019/0184566	12/2018	Yokoi	N/A	N/A
2019/0344446	12/2018	Goto et al.	N/A	N/A
2020/0027765	12/2019	Goto et al.	N/A	N/A
2020/0030973	12/2019	Goto et al.	N/A	N/A
2021/0178578	12/2020	Nakai et al.	N/A	N/A
2021/0252713	12/2020	Mimura	N/A	B25J 9/1697
2021/0257242	12/2020	Yoshida et al.	N/A	N/A
2023/0019019	12/2022	Kitano et al.	N/A	N/A
2024/0253233	12/2023	Hashizaki et al.	N/A	N/A
2024/0346641	12/2023	Kitano et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109564889	12/2018	CN	N/A
109716500	12/2018	CN	N/A
109789569	12/2018	CN	N/A
109791908	12/2018	CN	N/A
114830320	12/2021	CN	N/A
115039214	12/2021	CN	N/A
109564889	12/2022	CN	N/A

109716500	12/2022	CN	N/A
109791908	12/2022	CN	N/A
117581349	12/2023	CN	N/A
119072778	12/2023	CN	N/A
2006-120820	12/2005	JP	N/A
2014-128855	12/2013	JP	N/A
2017-052053	12/2016	JP	N/A
6111065	12/2016	JP	N/A
2018-051670	12/2017	JP	N/A
2018-051671	12/2017	JP	N/A
2018-056256	12/2017	JP	N/A
6718352	12/2019	JP	N/A
6741537	12/2019	JP	N/A
6741538	12/2019	JP	N/A
2021-091077	12/2020	JP	N/A
2021-091078	12/2020	JP	N/A
2023-030876	12/2022	JP	N/A
2023-163010	12/2022	JP	N/A
7522178	12/2023	JP	N/A
2024-138455	12/2023	JP	N/A
7582797	12/2023	JP	N/A
10-2019-0013863	12/2018	KR	N/A
10-2019-0058498	12/2018	KR	N/A
10-2019-0058500	12/2018	KR	N/A
10-2121972	12/2019	KR	N/A
10-2121973	12/2019	KR	N/A
10-2172266	12/2019	KR	N/A
10-2022-0137071	12/2021	KR	N/A
10-2024-0039020	12/2023	KR	N/A
201827184	12/2017	TW	N/A
201827185	12/2017	TW	N/A
201828390	12/2017	TW	N/A
1659491	12/2018	TW	N/A
1691391	12/2019	TW	N/A
1691392	12/2019	TW	N/A
202131438	12/2020	TW	N/A
202137381	12/2020	TW	N/A
202138138	12/2020	TW	N/A
1770712	12/2021	TW	N/A
1778436	12/2021	TW	N/A
202317335	12/2022	TW	N/A
I825943	12/2022	TW	N/A
202400385	12/2023	TW	N/A
2018/061957	12/2017	WO	N/A
2018/062153	12/2017	WO	N/A
2018/062156	12/2017	WO	N/A
2019/097793	12/2018	WO	N/A
2021/117619	12/2020	WO	N/A
2021/117620	12/2020	WO	N/A
2021/161582	12/2020	WO	N/A

Primary Examiner: Mazumder; Tapas

Attorney, Agent or Firm: METROLEX IP LAW GROUP, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. application Ser. No. 17/783,297, filed on Jun. 8, 2022, which is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2019-222900 filed on Dec. 10, 2019 and No. 2020-099458 filed on Jun. 8, 2020, the entire contents of all of which are incorporated herein by reference.

TECHNICAL FIELD

(1) This invention relates to a video confirmation computer for confirming a video of operation performed by a robot.

BACKGROUND ART

(2) Conventionally, robots have been used, for example, to transport wafers and other substrates. The robot of PTL 1 includes a controller for controlling the motion of the robot.

PRIOR-ART DOCUMENTS

Patent Documents

(3) PTL 1: Japanese Patent Application Laid-Open No. 2008-28134

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(4) The robot may generate an error for various reasons. In this case, maintenance work must be performed according to the error that occurs. In order to achieve a high level of information management in wafer processing facilities or the like, data transfer to another location or communication with an external network is often restricted. In this situation, there is a need to be able to quickly identify and resolve the cause of a robot error.

(5) The present invention is made in view of the circumstances described above, and an object of the present invention is to enable the operator to quickly and well understand the situation when, for example, a robot error occurs.

Means for Solving the Problems

(6) The problem to be solved by the present invention is as described above, and next, means for solving the problem and effects thereof will be described.

(7) According to an aspect of the present invention, a video confirmation computer having the following configuration is provided. That is, this video confirmation computer is used to confirm a video of operation performed by a robot. The video confirmation computer includes a storage unit and a processor. The storage unit is capable of storing information. The processor outputs information based on stored contents of the storage unit. The storage unit stores information of a position of an electric motor and information of the video. The electric motor drives a link body of the robot. The information of the position of the electric motor is received from the controller of the robot. The information of the video is obtained by a camera attached to the robot. The processor makes at least one of the video confirmation computer itself and a computer connected to the video confirmation computer display a model area and a video area side by side. In the model area, a two-dimensional or three-dimensional model reproducing a posture of the robot is displayed by computer graphics. In the video area, the video is displayed.

(8) The two areas are displayed side by side in such a way, therefore, the operator can understand the situation regarding the operation of the robot in a composite and intuitive manner. Thus, the operator can handle the situation smoothly and accurately.

Effects of the Invention

(9) According to the invention, for example, in the event of a robot error, the operator can grasp the situation quickly and well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a planar cross-sectional view showing a portion of a semiconductor processing facility to which a video confirmation computer according to an embodiment of the present invention is applied.

(2) FIG. 2 is a side cross-sectional view showing a portion of the semiconductor processing facility being cut.

(3) FIG. 3 is a schematic diagram illustrating a communication network constructed in a semiconductor processing facility.

(4) FIG. 4 is a diagram showing a display example of a status confirmation screen displayed on a video confirmation computer.

(5) FIG. 5 is a diagram illustrating a display example of a log analysis screen displayed on a handling computer.

(6) FIG. 6 is a diagram showing a display example of an error handling support screen.

(7) FIG. 7 is a diagram showing a display example of a manual screen.

EMBODIMENT FOR CARRYING OUT THE INVENTION

(8) Next, the embodiment of the invention will be described with reference to the drawings. FIG. 1 is a planar cross-sectional view showing a portion of a semiconductor processing facility 20 to which an embodiment of the present invention is applied. FIG. 2 is a side cross-sectional view showing a portion of the semiconductor processing facility 20 being cut. FIG. 1 and FIG. 2 show a robot 27 in various states of motion with two dotted chain lines.

(9) The semiconductor processing facility 20 applies predetermined processing to a wafer 24, which is the substrate to be processed. In this embodiment, the wafer 24 is a semiconductor wafer. Various processing can be applied to the wafer 24, such as heat process, impurity introduction process, thin film formation process, lithography process, cleaning process, or planarization process. Substrate processing other than the above-described substrate processing may be performed in the semiconductor processing facility 20.

(10) The semiconductor processing facility 20 includes a wafer processing device 22 and a wafer transfer device 23. The semiconductor processing facility 20 is a facility predetermined by, for example, the SEMI standard. SEMI is an abbreviation for the Semiconductor Equipment and Materials International. In this case, for example, a hoop 25 and a hoop opener 26 for opening and closing the hoop 25 follow the SEMI standards E47.1, E 15.1, E57, E62, E63, E84, or other specifications. However, the configuration of the semiconductor processing facility 20 may differ from the SEMI standard.

(11) In the wafer processing device 22, a processing space 30 that is filled with a predetermined gas is formed. The wafer processing device 22 applies the processing described above to the wafer 24 in the processing space 30. The wafer processing device 22 includes, in addition to a main body of the processing device that applies processing to the wafer 24, a processing space forming unit that forms the processing space 30, a wafer transportation unit that transports the wafer 24 in the processing space 30, and a regulating device that regulates the atmospheric gas filled in the processing space 30. The regulating device is realized by a fan filter unit or the like.

(12) The wafer transfer device **23** takes the wafer **24** before processing out from the hoop **25** and supplies it to the wafer processing device **22**. The wafer transfer device **23** takes the wafer **24** after processing out from the wafer processing device **22** and houses it into the hoop **25** again. The wafer transfer device **23** functions as a front-end module device (Equipment Front End Module; EFEM). In the semiconductor processing facility **20**, the wafer transfer device **23** serves as an interface unit that transfers the wafer **24** between the hoop **25** and the wafer processing device **22**. The wafer **24** is transferred between the space in the hoop **25** and the processing space **30** in the wafer processing device **22** while moving through a preparation space **29** with a high cleanliness level that is filled with a predetermined atmospheric gas.

(13) The preparation space **29** is a closed space where contamination control is performed. In the preparation space **29**, suspended microparticles in the air are controlled below a limited cleanliness level, and environmental conditions such as temperature, humidity, and pressure are controlled as necessary. In this embodiment, the processing space **30** and the preparation space **29** are maintained at a specified cleanliness level so as not to adversely affect the processing of the wafer **24**. For this cleanliness level, for example, the CLASS 1 specified by the ISO (International Organization for Standardization) is used.

(14) The robot **27** functions as a wafer transfer robot. In this embodiment, the robot **27** is realized by a SCARA type horizontal articulated robot. SCARA is an abbreviation for Selective Compliance Assembly Robot Arm. The robot **27** is located in the preparation space **29**.

(15) As shown in FIG. 2 and the like, the robot **27** includes a base **41**, a robot arm **42**, a vertical drive actuator **43a**, a horizontal drive actuator **43b**, and a controller **44**.

(16) The base **41** functions as a base member to support the robot arm **42**. The robot arm **42** is attached to the top surface of the base **41**.

(17) The robot arm **42** includes a link structure with a plurality of link bodies that are connected sequentially in a direction from a base end to a tip end. A robot hand **45** is provided at the tip end of the robot arm **42**. The robot hand **45** can hold and release the wafer **24**. There are various methods of holding the wafer **24** by the robot hand **45**, for example, placing the wafer **24** on top of the robot hand **45**, clamping the wafer **24** by the robot hand **45**, suctioning the wafer **24** to the robot hand **45** by negative pressure, and the like.

(18) The vertical drive actuator **43a** displaces the robot arm **42** in a vertical direction. The vertical drive actuator **43a** is configured as an electric motor, for example. By moving the robot arm **42** in the vertical direction, the height position of the robot hand **45** can be changed.

(19) The horizontal drive actuator **43b** rotates each of the link bodies of the robot arm **42** individually about corresponding joint axes. The horizontal drive actuator **43b** is configured as an electric motor, for example. Rotation of each link body about the vertical joint axis allows the robot hand **45** to move in a horizontal plane.

(20) The controller **44** controls the vertical drive actuator **43a** and the horizontal drive actuator **43b** to move the robot hand **45** to a predetermined position. The controller **44** includes, as shown in FIG. 3, a storage circuit **46**, a computing circuit **47**, and an output device **48**. The storage circuit **46** stores the predetermined program and various data. The computing circuit **47** performs arithmetic processing in accordance with the program. The output device **48** outputs control signals to the vertical drive actuator **43a** and the horizontal drive actuator **43b** based on the computing results of the computing circuit **47**. The storage circuit **46** is realized by, for example, a RAM, a ROM, and an HDD, etc. The computing circuit **47** is realized by, for example, a CPU.

(21) As shown in FIG. 3, the semiconductor processing facility **20** includes a higher-level controller **61**. The higher-level controller **61** transmits commands for executing the semiconductor processing process to various devices included in the semiconductor processing facility **20**.

(22) The higher-level controller **61** is wired or wirelessly connected to the controller **44** described above provided by the robot **27**. The higher-level controller **61** sends commands to the controller **44** so that the robot **27** performs the necessary operation at the necessary timing. The controller **44**

controls the robot **27** according to the commands received from the higher-level controller **61**.

(23) If any error occurs, the controller **44** transmits information regarding the error to the higher-level controller **61**. The program for the controller **44** to control the robot **27** includes a plurality of modules. Here, a module is a portion of the program for realizing a certain unit of operation by the robot **27**. The units of operation may include, for example, operation of placing the wafer **24** held by the robot hand **45** at a predetermined position, operation of holding the placed wafer **24** by the robot hand **45**, and the like, but not limited to.

(24) The vertical drive actuator **43a** and the horizontal drive actuator **43b** are each configured as an electric motor. The electric motor is of a known configuration and includes a motor drive circuit, a motor main body, a rotational position sensor, and a current sensor, which are not shown. The motor main body includes a stator, a rotor, and an output shaft. The controller **44** supplies current to a motor drive circuit through a power circuit so that the electric motor performs the desired operation. As a result, the output shaft of the electric motor rotates according to the flowing current.

(25) The current flowing in the motor drive circuit is determined by the target value of the operation of the electric motor and the detected value of the operation. The target value of the operation of the electric motor can be, for example, a target value for rotational position, rotational speed, or rotational acceleration. The detected value can likewise be, for example, a detected value for rotational position, rotational speed, or rotational acceleration. In controlling the current flowing in the motor drive circuit, control laws such as feedback control laws or adaptive control laws may be used.

(26) In the robot arm **42**, a camera **62** is fixed to the link body to which the robot hand **45** is fixed. As shown in FIG. **3**, this camera **62** is connected via a network to a video confirmation computer **65**. The video confirmation computer **65** can be, for example, a mobile terminal, a personal computer, or the like.

(27) The semiconductor processing facility **20** includes a camera **63** for taking a video of the robot **27**. The camera **63** is wired or wirelessly connected to the controller **44**.

(28) If some abnormality occurs in the robot **27** and a need for troubleshooting arises, the operator can operate some kind of computer to obtain information for handling. This computer may be referred to as a handling computer **70**. In this embodiment, the higher-level controller **61** serves as the handling computer **70**.

(29) The video confirmation computer **65** includes a storage circuit (storage unit) **66** and a computing circuit (processor) **67**. The storage circuit **66** is realized by, for example, a RAM, a ROM and an HDD, etc. The computing circuit **67** is realized by, for example, a CPU.

(30) For example, the information shown in (1) to (6) below is stored in the storage circuit **66** of the video confirmation computer **65** and the storage circuit **46** of the controller **44** (hereinafter simply referred to as the storage circuit). This storage operation may be performed continuously or discretely at intervals. (1) Information about the images captured by cameras **62** and **63** (2) Information about the signals received by the controller **44** from the higher-level controller **61** and the signals sent by the controller **44** to the higher-level controller **61** (communication log information) (3) Information about the program executed by the controller **44** to control the robot **27** (4) Information about the current value, position, speed, acceleration (including negative acceleration, the same hereinafter), position deviation, velocity deviation, and acceleration deviation of the electric motor provided in the robot **27** (5) Information about the output signals from various sensors (for example, light emitting/shading sensor, piston position sensor of pneumatic actuator, pressure sensor, valve position sensor of solenoid valve) provided in the robot **27** (6) Information about input signals to various devices (for example, solenoid valve) provided in the robot **27**

(31) Here, the current value of the electric motor is detected by a current sensor installed in the motor drive circuit. The position of the electric motor can be obtained based on the detection of the rotational position sensor installed in the electric motor. The speed and acceleration of the electric

motor can be obtained by differentiating the detected values of the rotational position sensor with time. The position deviation, speed deviation, and acceleration deviation can be obtained by calculating the difference between the above-described position, speed, and acceleration and the target position, target speed, and target acceleration.

(32) A web server application and a database application are previously installed in the controller **44** and the program is stored in the storage circuit **46** described above.

(33) By the way, for some reason, an error may occur in the robot **27**. In this case, the video confirmation computer **65** can display a status confirmation screen **71** as shown in FIG. **4**, based on contents stored in the storage circuit **66**. This status confirmation screen **71** includes a model area **72**, a video area **73**, a log area **74** and a graph area **75**.

(34) In the model area **72**, a two-dimensional or three-dimensional model reproducing the posture of the robot **27** at a certain point in time is displayed by computer graphics. The posture of the displayed model is calculated based on the positions of the motors stored in the storage circuit **66**. A seek bar **72a** is located at the bottom of the model area **72**. In this seek bar **72a**, a slider, a play button, a pause button, and a reverse play button are arranged. This allows, in the display of the model area **72**, the operation of the robot **27** to be reproduced or stopped at a certain point in time. Even when, for example, the real robot **27** is in a position where it is hidden from the camera **63** and cannot be captured, the operator can easily understand what posture the robot **27** was in by referring display contents on the model area **72**.

(35) In the video area **73**, the captured image obtained by the camera **62** and stored in the storage circuit **66** is displayed. The play position of the image can be specified by the seek bar **72a**. Thus, the reproduction of the model operation in the model area **72** and the play of the video in the video area **73** can be synchronized.

(36) In the log area **74**, a communication log stored in the storage circuit **66** is displayed. Information about the program executed to control the robot **27** may be displayed on the log area **74**.

(37) In the graph area **75**, information about the electric motor current value, position, speed, acceleration, position deviation, speed deviation, and acceleration deviation stored in the storage circuit **66** are displayed in different colored graphs. In the example shown in FIG. **4**, a line graph is displayed with each value on the vertical axis and time on the horizontal axis. A vertical straight line (time graphic) is displayed on each graph in the graph area **75**. The position of this line corresponds to the time of the moment displayed on the model area **72** and the video area **73**.

(38) When some error occurs in the robot **27**, the information stored in the storage circuit **66** within a predetermined time range including the time of the error (for example, from a several seconds before the time of the error to a several seconds after the time of the error) is the target of display on the status confirmation screen **71**.

(39) When the play button is pressed on the seek bar **72a**, the information to be displayed is played on the model area **72** and the video area **73**. Similarly, when the reverse play button is pressed, the information to be displayed is played in reverse. When the pause button is pressed, the play/reverse play is paused, and when the pause button is pressed again, the pause is released. The seek bar **72a** may include buttons for performing known fast forward/rewind operation.

(40) When the information to be displayed is played, the reproduced motion of the robot **27** is displayed on the model area **72**, and the captured images by the camera **62** are displayed dynamically on the video area **73**. During play, the communication log is displayed on the log area **74** and a graph is displayed on the graph area **75**. In the log area **74**, the log history at the time indicated by the seek bar **72a** is displayed. This log history is appended moment by moment in synchronization with play. In the graph area **75**, the above-described time graphic in each graph moves horizontally in synchronization with play.

(41) The above configuration allows, when an error occurs, the operator to quickly identify and handle the cause of the error from various perspectives by operating the video confirmation

computer **65**.

(42) In the above example, the information stored in the storage circuit **66** in the predetermined time range before and after the error occurred is the target for display on the status confirmation screen **71**. Alternatively, the information stored for the unit of operation or the module that was being executed in the controller **44** at the time the error occurred may be the target for display on the status confirmation screen **71**.

(43) The model area **72** and the video area **73** are arranged side by side in the same status confirmation screen **71** so that they are in close proximity to each other. Thus, the information in the two areas allows the operator to understand the situation in which the error occurred in a composite and intuitive manner.

(44) The overall situation of the robot **27** is obtained from the model area **72**, and the detailed situation around the tip of the robot hand **45** is obtained from the video area **73**. Thus, the information of the two-dimensional or three-dimensional model displayed on the model area **72** and the information of the captured video displayed on the video area **73** can be complementary to each other. In this sense, it is advantageous for both video to be displayed side by side.

(45) In addition, the appending of the log history in the log area **74** is done at a timing that matches the time related to the display on the model area **72** and the video area **73**. Therefore, there is no confusion for the operator in understanding the situation.

(46) Since the model area **72** and the video area **73** are arranged adjacent to each other, it is easier for the operator to grasp the model video and the captured image as integral information. Therefore, it is possible to realize a configuration which does not cause discomfort even if the video in the video area **73** is played/stopped by the seek bar **72a** in the model area **72**.

(47) In the example of FIG. **4**, the model area **72** and the video area **73** are arranged so that the model area **72** is on the left and the video area **73** is on the right. However, the arrangement may be reversed to the left and right. Also, the model area **72** and the video area **73** can be arranged side by side vertically.

(48) Next, with reference to FIG. **5**, the log analysis screen **81** that the controller **44** can be displayed on the display of the handling computer **70** (for example, a display **68** shown in FIG. **3**) will be described.

(49) When some error occurs in the robot **27**, the communication log describing the contents of the communication before and after that point in time is highly valuable information for identifying the cause of the error, etc. However, the communication log generally include both those that have high relevance to the error and those that have low relevance to the error, and in many cases, those that have low relevance to the error make up the majority of the communication log. Thus, those with high relevance to the error may be buried elsewhere, making them difficult to find.

(50) Accordingly, in this embodiment, in the log analysis screen **81** that the controller **44** outputs to the handling computer **70**, has a function which extracts and displays items that are highly relevant to the error among items in the communication log. This facilitates the identification of the cause of the error and its handling. The handling computer **70** functions as a terminal to the controller **44**.

(51) Specifically, a communication code indicating the type of communication is described for each line of the communication log. In the storage circuit **46** of the controller **44**, information of communication codes that are highly relevant to errors in the communication log is previously stored so that the information of communication codes is in association with information identifying the type of error (for example, error code). When extraction of the communication log is indicated by the operator, the communication log is filtered and output from the controller **44** to the handling computer **70**, so that only those types of the communication log that are highly relevant to the error code are displayed.

(52) When the controller **44** displays the communication log in such a filtered form on the handling computer **70**, for example, the following (1) through (5) may be displayed at the same time. This log analysis screen **81** allows the operator to more easily understand the situation. (1) An error

description area **82**. Display contents in the error description area **82** include, for example, the date and time of the error, the error code, and a summary of the error. (2) A detailed log area **83**. Display contents in the detailed log area **83** include the communication log extracted as described above from a predetermined time before the error occurred to the point in time when the error occurred. (3) An operation position area **84**. The operating position area **84** can display numerical values indicating the posture of the robot **27** at the time the error occurred, in the form of a detected position or target value of the electric motor. (4) A program status area **85**. The program status area **85** displays information for identifying the module that was executed to control the robot **27** according to the communication log displayed on the detailed log area **83**. However, each line of the communication log may be selectable in the detailed log area **83**, and the program corresponding to the selected line of the log may be displayed on the program status area **85**. (5) An input/output signal area **86**. The input/output signal area **86** displays output signals from various sensors that the robot **27** has and input signals to various devices.

(53) Next, the error handling support screen **88**, which can be displayed by the handling computer **70**, will be described with reference to FIG. **6**.

(54) The storage circuit **46** of the controller **44** stores data indicating how to handle with the case that some error occurs in the robot **27**. This handling method is output from the controller **44** to the handling computer **70** and displayed on a display (for example, the display **68** shown in FIG. **3**) to assist the operator in dealing with the error.

(55) Contents displayed on the display **68** to assist the work in handling the error can be, for example, the following (1) through (3). (1) An error description area **89**. The error description area **89** displays the date and time the error occurred, the error code, and the like. In the error description area **89**, errors that occurred within a predetermined past time from the time of the most recent error occurrence or errors that occurred a predetermined number of times in the past from the time of the most recent error occurrence may be selectively displayed. Hereafter, the past errors displayed on the error description area **89** may be referred to as the error history. (2) A handling contents description area (detailed description area) **90**. In the handling contents description area **90**, the handling method corresponding to the error that is the target of description in the error description area **89** is displayed in detail. The handling method is previously created corresponding to the error code by the manufacturer of the robot **27**, and is stored in the storage circuit **46** of the controller **44** in the form of appropriate electronic data such as HTML data, image data, video data, PDF data, etc. When an error history is selected in the error description area **89**, a method of handling corresponding the error history may be displayed on the handling contents description area **90**. The video data can be, for example, data captured by the camera **63** of the robot **27** when an error occurred in the past. (3) A related items list area **91**. The related items list area **91** displays a list of items related to the error described in the error description area **89**. When an error history is selected in the error description area **89**, the items related to that error history may be displayed on the related items list area **91**. The list consists of one or more related items. The related items can be, for example, the following (a) or (b). (a) A record of work performed in the past in response to the current error or the selected error history. When an error occurs, the operator can make the work record creation screen be displayed on the display **68** by operating the appropriate user interface device. This interface can be, for example, a touch panel integrated with the display **68**, or hardware keys located near the display **68**, but however, it is not limited to. On the work record creation screen, the operator can create a work record by specifying the error code, error occurrence date and time, work date and time, operator, contents of work and results, title of the work record, importance of the work record, or the like. The work record which the operator inputs into the handling computer **70** is transmitted from the handling computer **70** to the controller **44**. Upon receiving the work record data, the controller **44** stores the work record in the storage circuit **46** in association with the error code. In the related items list area **91**, for example, the importance and title of the work record are displayed. (b) An error associated with the current error or the

selected error history. The associated error includes, for example, other errors that may occur simultaneously with an error and other errors that occur as a result of the occurrence of an error, but is not limited to. Associations between errors are created previously by the manufacturer of the robot **27** and stored in the storage circuit **46** of the controller **44**. Associations between errors may be configured to be registered in the storage circuit **46** by operation of the operator. The related items list area **91** displays, for example, the error code.

(56) By operating the user interface device described above as appropriate, the operator may select the related item displayed in the related items list area **91** as appropriate. Information on the selected related item is transmitted from the handling computer **70** to the controller **44**. The controller **44** transmits various data to the handling computer **70** so that, when a work record is selected, the contents of that work record are displayed in the handling contents description area **90**, and when an associated error is selected, how to handle that error is displayed in the handling contents description area **90**.

(57) Although a plurality of related items can be displayed in the related items list area **91**, the controller **44** can control the handling computer **70** to sort and display the related items so that the related item with the highest priority is at the top and the related item with the lowest priority is at the bottom. FIG. **6** shows the related items sorted according to priority in the related items list area **91**. The priority can be determined based on, for example, the frequency of occurrence of the error, the frequency with which the description screen is displayed, the importance specified in the work record, and the relevance determined from the current sensor and position sensor detection values when the error occurs. If multiple perspectives are used to determine the priority, the controller **44** can, for example, calculate the sum of a numerical index of the frequency of error occurrence, etc. multiplied by an appropriate weight, and the resulting sum can be used as the priority.

(58) For example, when a collision occurs, an abnormality such as an increase in position deviation is detected simultaneously for multiple motors which drive joints. On the other hand, for example, if the cable of a certain motor is disconnected, an abnormality will be detected only for that motor. The system may store such assumed abnormality detection patterns in advance, calculate the similarity between the actual abnormality detection pattern and the assumed abnormality detection pattern when an abnormality occurs, and display in priority the troubleshooting contents for the abnormality corresponding to the assumed abnormality detection pattern with the high similarity. As the abnormality detection pattern, a combination of occurrence/non-occurrence of multiple detection items (position deviation, speed deviation, etc.) may be used, or detected values such as the current value of a certain motor when an abnormality occurs may be used. For example, when a combination of occurrence/non-occurrence of multiple detection items is used as an abnormality detection pattern, the number of occurrence/non-occurrence combinations of multiple detection items that match may be used as the similarity. In this case, some detection items may be weighted to increase the effect on the similarity of important detection items.

(59) The above configuration makes it easier to identify and handle the cause of the error. In addition, by making the work records registered by the user also the target of display on the error handling support screen **88**, it becomes possible to handle errors based on the accumulation of knowledge acquired through past handling, and such knowledge can be shared by multiple persons. Related items are displayed in order of priority, so that errors that have occurred can be handled efficiently.

(60) The storage circuit **46** of the controller **44** stores methods for identifying the cause of the error, documents describing how to handle with the error, and a manual describing how to use the robot. By operating the user interface device according to the situation, the user can make these documents and manuals displayed on the display **68** for reference, as shown in FIG. **7**.

(61) A manual screen **93** is briefly described below. On the left side of the manual screen **93**, a table of contents display area **94** is located. In the table of contents display area **94**, the items of the table of contents of the manual are displayed in list form. On the right side of the manual screen **93**, a

detailed display area **95** is located. In this detailed display area **95**, the specific contents of the manual corresponding to the item of the table of contents selected in the table of contents display area **94** can be displayed. A search box **96** is located above the table of contents display area **94**, allowing text searches to be performed on text data of the manual. This eliminates the need to bring the paper manual closer to the robot **27**.

(62) The display of each of the log analysis screen **81**, the error handling support screen **88**, and the manual screen **93** is realized by the collaboration of the web server application previously installed in the controller **44** and the web browser application previously installed in the handling computer **70**. By starting a web browser and operating the web browser on the handling computer **70**, the above-mentioned log analysis screen **81**, the error handling support screen **88**, and the manual screen **93** can be displayed on the display **68**, for example, in the form of web pages or the like.

(63) The handling computer **70** for viewing the various screens (in other words, a terminal to the controller **44**) can be prepared separately from the higher-level controller **61**. In any case, a web browser application is installed on the viewing computer. The web server of the controller **44** determines the contents for display from the contents which are stored in the storage circuit **46** in response to a request from the web browser of the viewing computer, and transmits various data such as HTML data. The web browser on the viewing computer renders a screen such as a web page based on the various data received. In this configuration, the operator only needs to prepare an appropriate terminal with a general web browser application installed as the viewing computer, and the functions of the log analysis screen **81**, the error handling support screen **88**, and the manual screen **93** can be used. Thus, in this embodiment, there is no need to install special software such as a dedicated application on the terminal. Therefore, this configuration is suitable for semiconductor manufacturing plants, etc., where the bringing in of electronic equipment is generally strictly restricted and a high level of confidentiality measures are required for the electronic equipment brought in.

(64) As explained above, in this embodiment, the video confirmation computer **65** for confirming the video related to the operation of the robot **27**, includes the storage circuit **66** and the computing circuit **67**. The storage circuit **66** is capable of storing information. The computing circuit **67** outputs information based on the stored contents of the storage circuit **66**. The storage circuit **66** stores information of the position of the electric motor that drives the link body of the robot **27** and information of the video. The information of the position of the electric motor is received from the controller **44** of the robot **27**. The information of the video is obtained by the camera **62** attached to the robot **27**. The computing circuit **67** makes the display provided with the video confirmation computer **65** itself display the model area **72** and the video area **73** side by side. In the model area **72**, a two-dimensional or three-dimensional model reproducing the posture of the robot **27** is displayed by computer graphics. In the video area **73**, a video is displayed.

(65) The two areas are displayed side by side in such a way, therefore, the operator can understand the situation regarding the operation of the robot **27** in a composite and intuitive manner. Thus, the operator can smoothly and accurately handle the situation (for example, occurrence of an error).

(66) In the video confirmation computer **65** of this embodiment, the time corresponding to the model displayed in the model area **72** and the time corresponding to the video displayed in the video area **73** are synchronized.

(67) This allows the reproduction of the model operation in the model area **72** and the play of the video in the video area **73** to be synchronized. Thus, the operator viewing the screen can easily grasp the situation.

(68) In the video confirmation computer **65** of this embodiment, the storage circuit **66** also stores the communication log in which the controller **44** communicated with other devices. In addition to the model area **72** and the video area **73**, the computing circuit **67** also makes the log area **74** where the history of the communication log is output be displayed on the video confirmation computer **65** itself. The timing of displaying the history of the communication log in the log area **74** is

synchronized with the time in the model area 72 and the video area 73.

(69) As a result, changes in the display of the log area 74 are timed to be consistent with the times represented by the model area 72 and the video area 73. Thus, the operator is not confused in understanding the situation.

(70) In the video confirmation computer 65 of this embodiment, the common seek bar 72a for specifying the time corresponding to the model displayed in the model area 72 and for specifying the time corresponding to the video displayed in the video area 73 is displayed by the computing circuit 67.

(71) This allows the operator to specify, by intuitive operation using the seek bar, at which time the situation is to be displayed in the model area 72 and the video area 73. Thus, the operator can easily grasp the situation at the time he/she wants to know. In addition, since the seek bar 72a is common, a simplified screen can be realized that is less confusing during operation.

(72) While some preferred embodiments of the present invention have been described above, the foregoing configurations may be modified, for example, as follows.

(73) The information in the normal state of the robot 27 may be stored in at least one of the storage circuit 66 of the video confirmation computer 65 and the storage circuit 46 of the controller 44. In this case, when an error occurs, the information in the normal state can be displayed for reference in the video confirmation computer 65 or in the handling computer 70. This configuration is particularly advantageous in the graph area 75 in the status confirmation screen 71 displayed by the video confirmation computer 65. This is because the graph in the graph area 75 can vary greatly even in normal state, and it is difficult to distinguish between abnormal and normal conditions without a basis for comparison. Parameters that characterize the graph in normal state (for example, average value, maximum value, minimum value, or the like) can also be displayed in the graph in the graph area 75.

(74) The computer displaying at least any of the log analysis screen 81, the error handling support screen 88, and the manual screen 93 may be a teach pendant that is operated to perform teaching operation on the robot 27.

(75) Images obtained by the camera 62 attached to the robot 27 may also be used by the teach pendant to assist in performing the teaching operation.

(76) The video confirmation computer 65 and the handling computer 70 may be realized in one hardware (computer).

(77) The status confirmation screen 71 displayed by the video confirmation computer 65 may be displayed in the same browser-based manner as the log analysis screen 81 or the like.

(78) The video confirmation computer 65 can display the status confirmation screen 71 on the display of another computer connected to the video confirmation computer 65 instead of, or in addition to, the display of the video confirmation computer 65 itself.

DESCRIPTION OF THE REFERENCE NUMERALS

(79) 27 Robot 44 Controller (robot controller) 65 Video confirmation computer 66 Storage circuit (storage unit) 67 Computing circuit (processor) 68 Display (display unit) 72 Model area 73 Video area 74 Log area

Claims

1. A system for confirming a condition of a substrate shown in a video of operation performed by a robot, the system comprising: a camera attached to the robot; a storage unit capable of storing information; a processor that outputs information based on stored contents of the storage unit; and a display, wherein the camera is fixed to a link body to which a robot hand capable of holding the substrate is fixed, the storage unit stores: information of a position of an electric motor that drives the link body of the robot; and information of the video obtained by the camera, the information of the video configured to show the condition of the substrate, and the processor causes the display to

display the video.

2. The system according to claim 1, wherein the processor further causes the display to display a video area in which the video is displayed, and display a model area in which a two-dimensional or three-dimensional model reproducing a posture of the robot by computer graphics is displayed.
 3. The system according to claim 1, wherein the storage unit further stores a communication log comprising information about communication between the controller and another device, and the processor further causes the display to display a video area in which the video is displayed, and display a log area in which a history of the communication log is displayed.
 4. A method for confirming a condition of a substrate shown in a video of operation performed by a robot, the method comprising: storing in a storage unit: information of a position of an electric motor that drives a link body of the robot received from a controller of the robot; and information of the video obtained by a camera attached to the robot, the camera being fixed to the link body to which a robot hand capable of holding the substrate is fixed, the information of the video configured to show the condition of the substrate; and displaying the video on a display based on stored contents of the storage unit.
 5. A non-transitory computer-readable medium storing a program for confirming a condition of a substrate shown in a video of operation performed by a robot, the program including instructions that causes a processor to: store in a storage unit: information of a position of an electric motor that drives a link body of the robot received from a controller of the robot; and information of the video obtained by a camera attached to the robot, the camera being fixed to the link body to which a robot hand capable of holding the substrate is fixed, the information of the video configured to show the condition of the substrate; and display the video on a display based on stored contents of the storage unit.
 6. A system for confirming a condition of a substrate shown in a video of operation performed by a robot, the system comprising: a camera attached to the robot; a storage unit capable of storing information; a processor that outputs information based on stored contents of the storage unit; and a display, wherein the camera is fixed to a link body to which a robot hand capable of holding the substrate is fixed, the storage unit stores information of the video obtained by the camera, the information of the video configured to show the condition of the substrate, and the processor causes the display to display the video.
 7. A method for confirming a condition of a substrate shown in a video of operation performed by a robot, the method comprising: storing in a storage unit information of the video obtained by a camera attached to the robot, the camera being fixed to the link body to which a robot hand capable of holding the substrate is fixed, the information of the video configured to show the condition of the substrate; and displaying the video on a display based on stored contents of the storage unit.
 8. A non-transitory computer-readable medium storing a program for confirming a condition of a substrate shown in a video of operation performed by a robot, the program including instructions that causes a processor to: store in a storage unit information of the video obtained by a camera attached to the robot, the camera being fixed to the link body to which a robot hand capable of holding the substrate is fixed, the information of the video configured to show the condition of the substrate; and display the video on a display based on stored contents of the storage unit.
-